ENEE 132 — Basic Electronics I (Semiconductor Devices)

Lecture 3 - The Diode: Operation, Characteristics & Models

16 June 2026

Today's Agenda — Lecture Notes pp. 28–46

- **Part 1 — Diode Operation**
  - Diode structure and the schematic symbol
  - Forward bias, reverse bias, and reverse breakdown
- **Part 2 — The V-I Characteristic Curve**
  - The forward and reverse characteristic, and the complete curve
  - Dynamic resistance and temperature effects
- **Part 3 — Diode Models**
  - The ideal, practical, and complete models for circuit analysis
- **Part 4 — Wrap-Up**
  - Synthesis, key takeaways, and a look ahead to rectifiers

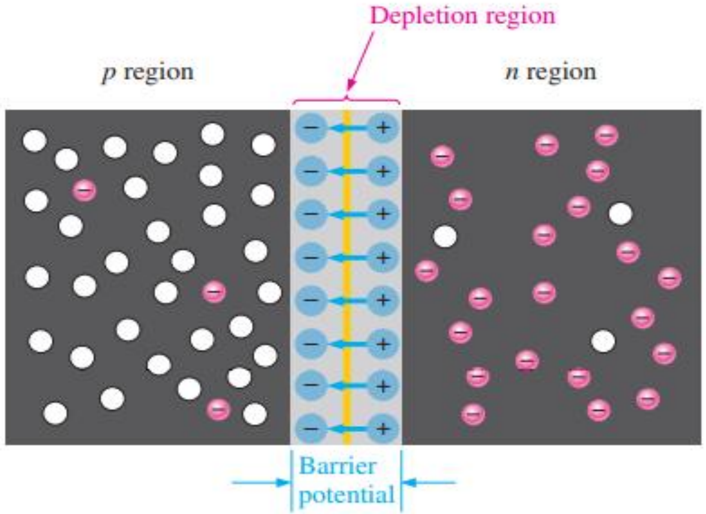
Today's Learning Objectives

- **By the End of This Lecture, You Will Be Able To**
  - Identify a diode's terminals (anode, cathode) and read its schematic symbol
  - Explain what happens to the depletion region and barrier potential under forward and reverse bias
  - Describe reverse breakdown and the breakdown voltage
  - Interpret the diode V-I curve — the knee, dynamic resistance, and temperature effects
  - Apply the ideal, practical, and complete models to analyze a simple diode circuit

Part 1 — Diode Operation

Where We Left Off — The PN Junction

- **Recap from Last Week**
  - If a piece of intrinsic silicon is doped so that part is n-type and the other part is p-type, a pn junction forms
  - Diffusion creates a depletion region of fixed ions at the junction
  - This sets up a barrier potential (≈ 0.7 V for silicon) that halts further diffusion at equilibrium



(b) For every electron that diffuses across the junction and combines with a hole, a positive charge is left in the *n* region and a negative charge is created in the *p* region, forming a barrier potential. This action continues until the voltage of the barrier repels further diffusion. The blue arrows between the positive and negative charges in the depletion region represent the electric field.

Fig 1.19(b) — Formation of the depletion region & barrier potential

Today’s Focus

- **From Last Week's Junction to a Working Device**
  - Connect the pn junction to a DC voltage source — this is called biasing
  - Forward bias makes it conduct; reverse bias almost completely blocks current
  - That one-way behaviour is the diode — our first electronic device
- **What We Will Cover Today**
  - How the diode operates under forward and reverse bias
  - Its voltage–current (V-I) characteristic curve
  - Three circuit models — ideal, practical, and complete

The Diode: Structure & Symbol

- **A Two-Terminal Device**
  - A diode is a two-terminal semiconductor device formed by two doped regions of silicon separated by a pn junction
  - The p-region terminal is the anode (A); the n-region terminal is the cathode (K)
  - Also called a general-purpose, rectifier, or signal diode, depending on its use
- **The Schematic Symbol**
  - The arrowhead represents the anode; the bar represents the cathode
  - The arrow points in the direction of conventional (forward) current flow
  - Diodes come in many packages, from through-hole to surface-mount

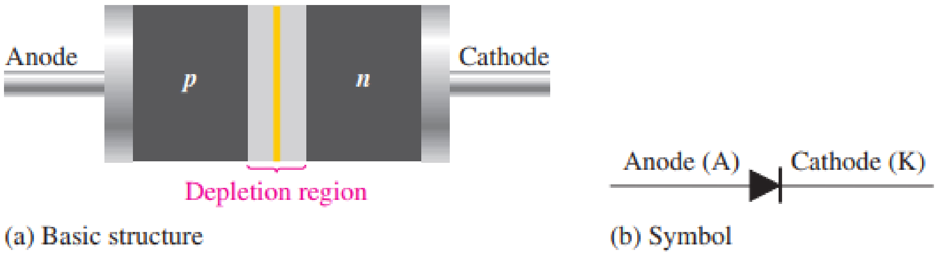


Fig 2.1 — The diode: basic structure and schematic symbol

Diode Packaging & Terminal Identification

- **Common Package Types**
  - Through-hole packages: the leads pass through holes in the circuit board
  - Surface-mount packages: SOD and SOT have gull-wing leads; SMA has L-shaped leads that bend under the package
- **Finding the Cathode**
  - Usually marked by a band, a tab, or another feature; on case-mounted types, the case itself is the cathode
  - In a single-diode SOT package, pin 1 is the anode and pin 3 is the cathode
  - A SOT package may contain one or two diodes — always check the datasheet to confirm the pinout

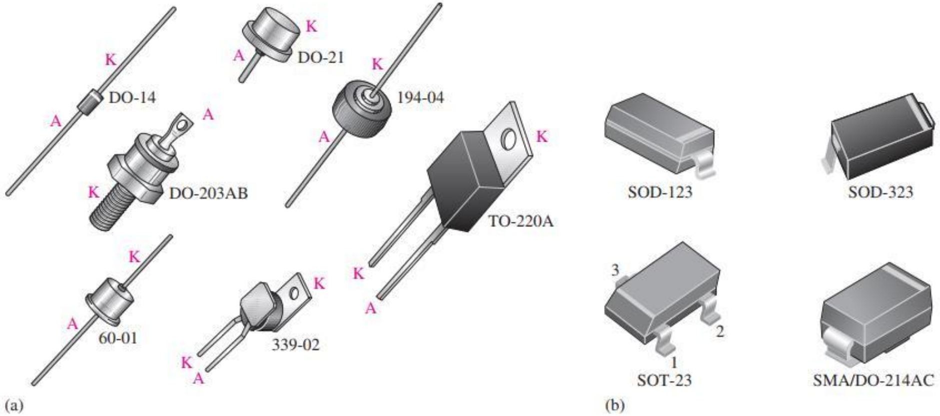


Fig 2.2 — Typical diode packages with terminal identification

Forward Bias — The Connection

- **How to Forward-Bias a Diode**
  - Connect the DC voltage source's positive terminal to the anode (p), negative to the cathode (n)
  - The bias voltage (VBIAS) must exceed the barrier potential (VB) (≈ 0.7 V for silicon)
  - Always include a current-limiting resistor in series
- **Things to Note**
  - The source then pushes majority carriers toward the junction
  - The diode is ON

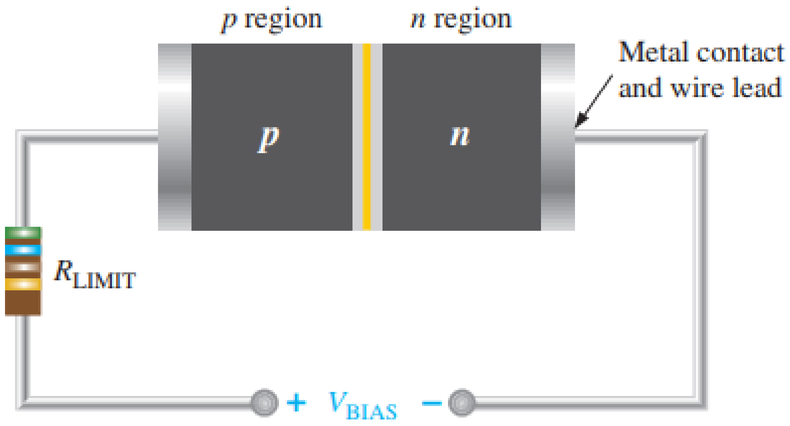


Fig 2.3 — A diode connected for forward bias

Forward Bias — What Happens at the Junction

- **At the Junction**
  - Majority carriers are pushed toward the junction (electron current), narrowing the depletion region
  - The barrier potential is reduced, so carriers cross easily
- **The Result**
  - A large forward current flows — the diode is ON
  - While conducting, the voltage across a silicon diode stays near 0.7 V

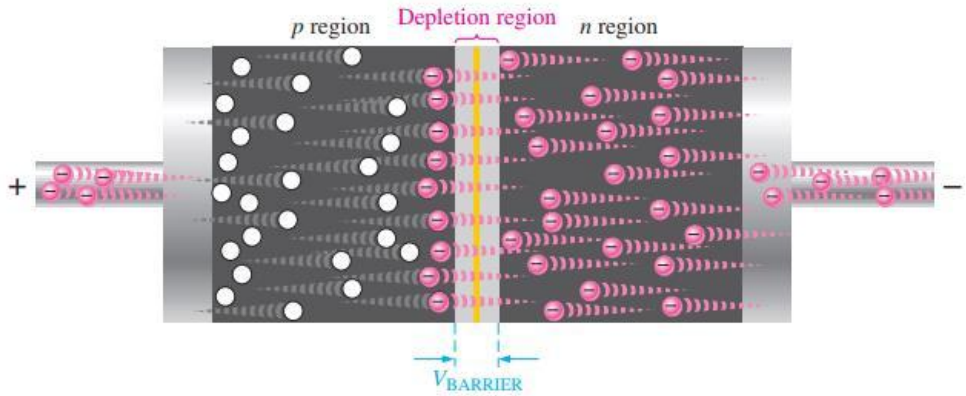


Fig 2.4 — Forward bias: majority-carrier flow and the barrier-potential voltage

Forward Bias — Hole Current in the p Region

- **Electrons Arrive, Then Travel as Holes**
  - After crossing the junction, electrons lose energy and drop into the valence band, combining with holes in the p region
  - The positive terminal pulls these valence electrons leftward; they hop from hole to hole toward the external lead
- **Hole Current**
  - As the electrons hop left, the holes effectively move right toward the junction — this is the hole current
  - The holes provide the only pathway for the valence electrons; a continuous supply of holes meets the incoming electrons

Forward Bias — Depletion Narrows & the 0.7 V Drop

- **The Depletion Region Narrows**
  - Electrons pushed in from the n side combine with holes on the p side, shrinking the depletion region
- **Climbing the Energy Hill**
  - The source gives electrons enough energy to overcome the barrier potential and cross the junction
  - Crossing costs each electron an energy equal to the barrier → i.e., 0.7 V drop appears across the junction
  - A small extra voltage drop appears across the p and n regions due to the internal resistance of the semiconductor material
  - This resistance is called the dynamic resistance and usually negligible

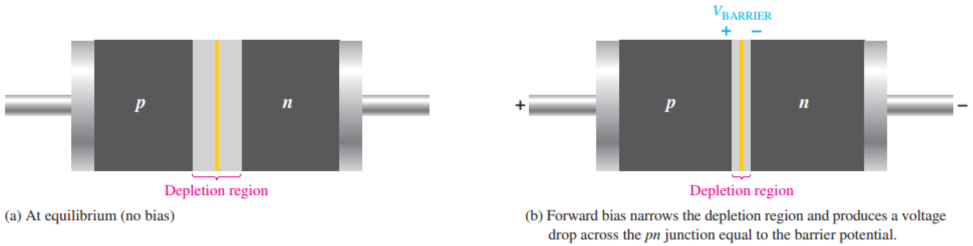


Fig 2.5 — The depletion region narrows; a 0.7 V drop appears across the junction

Reverse Bias — The Connection

- **How to Reverse-Bias a Diode**
  - Connect the DC voltage source's positive terminal to the cathode (n), negative to the anode (p)
  - A limiting resistor is not needed — there is essentially no current to limit
- **Things to Note**
  - The depletion region becomes much wider than in forward bias or at equilibrium (no bias)
  - The diode is effectively OFF

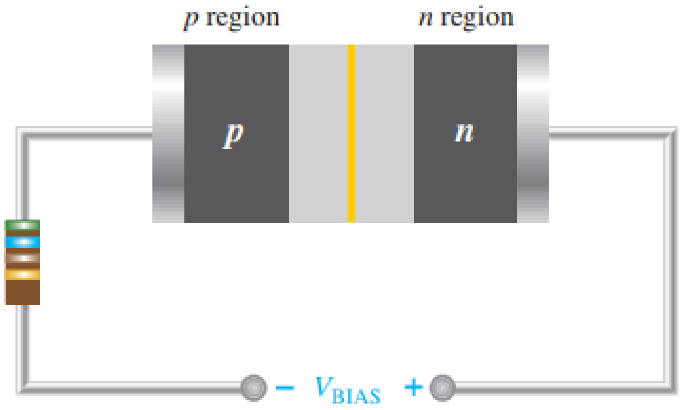


Fig 2.6 — A diode connected for reverse bias

Reverse Bias — What Happens

- **Carriers Pulled Away from the Junction**
  - The positive terminal pulls the n region's free electrons away from the junction; the negative terminal pulls the p region's holes away
  - This widens the depletion region and exposes more fixed ions
- **A Brief Transition, Then Stillness**
  - This is only a short transition current right after the voltage is applied
  - As the depletion region widens, its field grows until it equals VBIAS and the transition current stops

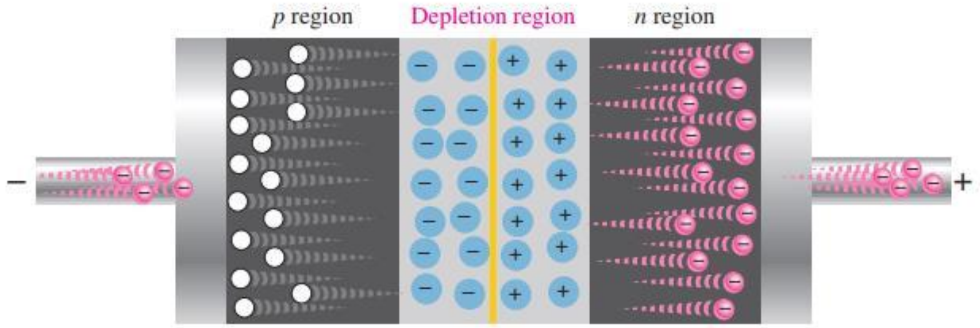


Fig 2.7 — The diode during the short transition after reverse bias is applied

Reverse Current — The Minority Carriers

- **Where the Tiny Current Comes From**
  - After the brief transition, a very small reverse current remains, carried by minority carriers from thermally generated electron–hole pairs
  - Minority electrons in the p region drift to the junction, 'fall down the energy hill', and cross easily — needing no extra energy
- **How Small?**
  - Typically microamps or nanoamps — small enough to neglect in most work

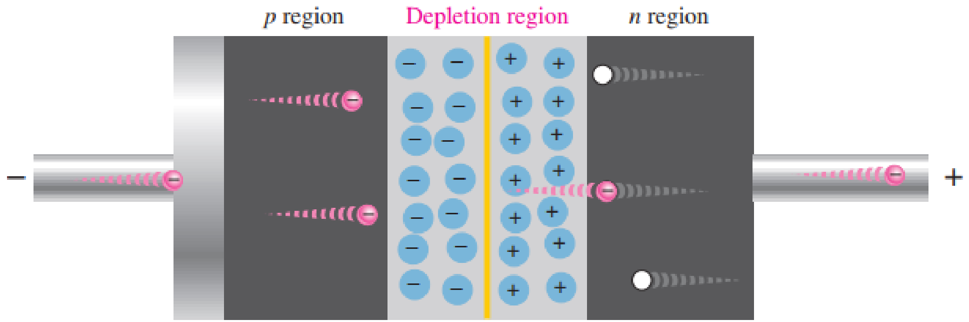


Fig 2.8 — The tiny reverse current from thermally generated minority carriers

Reverse Breakdown

- **Pushing Reverse Bias Too Far**
  - Raise the reverse voltage enough and the diode reaches its breakdown voltage (VBR)
  - Minority carriers are accelerated hard enough to knock others loose and the reverse current rises sharply
  - The multiplication of conduction electrons is known as the avalanche effect
- **How Large Is VBR?**
  - Set by the doping level, which the manufacturer controls
  - A typical rectifier diode: greater than 50 V; some specialized diodes: only about 5 V
- **Does It Damage the Diode?**
  - For an ordinary diode, unlimited breakdown current causes permanent damage — keep normal operation below VBR
  - This effect can be limited by connecting a current limiting resistor in series
  - Some diodes (e.g. the Zener) are built to operate in breakdown

Summary — Forward vs Reverse Bias

- **Forward Bias**
  - Source: + to anode, – to cathode (with a limiting resistor); VBIAS must exceed ≈ 0.7 V
  - Depletion region narrows → a large majority-carrier current flows (diode ON), ≈ 0.7 V across it
- **Reverse Bias**
  - Source: + to cathode, – to anode; the depletion region widens
  - Only a tiny minority-carrier leakage current flows known as reverse current (diode OFF) — until breakdown

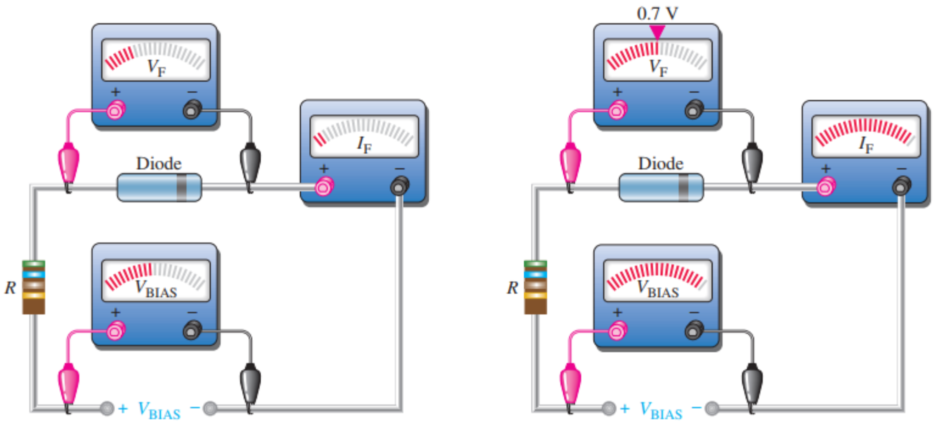
Quick Check #1 — Diode Operation

- **Question 1**
  - Name the two terminals of a diode and the region each connects to.
- **Question 2**
  - For forward bias, how is the source connected, and what condition must the voltage meet?
- **Question 3**
  - What happens to the depletion region and barrier potential under reverse bias?
- **Question 4**
  - Why does only a tiny current flow in a reverse-biased diode?
- **Question 5**
  - What is reverse breakdown, and why should an ordinary diode avoid it?

Part 2 — The V-I Characteristic of a Diode

Forward Bias — Taking the Measurements

- **Sweeping the Bias Voltage**
  - A resistor is used to limit the forward current to a safe (no damage or overheating of diode); part of VBIAS is dropped across it
  - From 0 V, gradually raise VBIAS and watch the diode voltage (VF) and the forward current (IF)
- **What the Meters Show**
  - Below ≈ 0.7 V: IF is very small and VF rises with VBIAS
  - At ≈ 0.7 V: IF climbs rapidly while VF barely changes
  - VF increases slightly due to the dynamic resistance which is the internal resistance of the semiconductor material



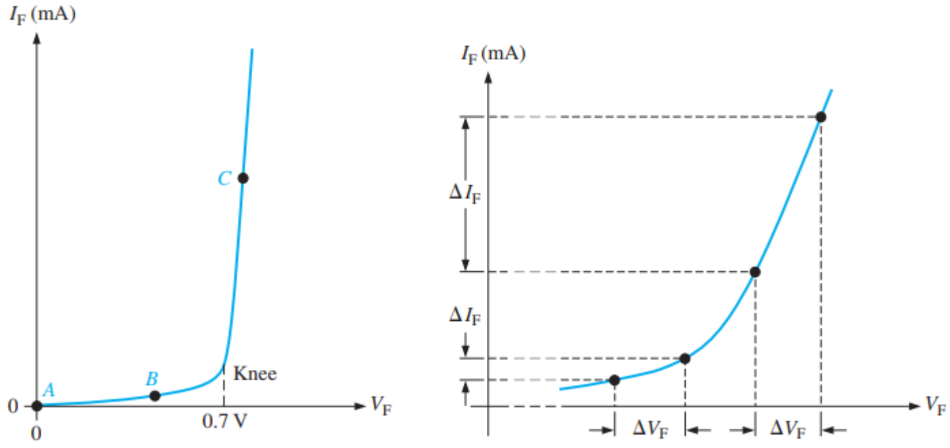
(a) Small forward-bias voltage ($V_F < 0.7$ V), very small forward current.

(b) Forward voltage reaches and remains nearly constant at approximately 0.7 V. Forward current continues to increase as the bias voltage is increased.

Fig 2.9 — Forward-bias measurements: VF and IF as VBIAS increases

The Forward V-I Curve & the Knee

- **Plotting Current vs. Voltage**
  - On the V-I curve, the horizontal axis is forward voltage (VF), the vertical axis is forward current (IF)
- **The Shape**
  - From 0 V, almost no current flows until VF approaches the barrier potential
  - At the knee (≈ 0.7 V for silicon), current begins to rise very steeply
  - Above the knee, small voltage increases give large current increases



(a) *V-I* characteristic curve for forward bias.

(b) Expanded view of a portion of the curve in part (a). The dynamic resistance r'_d decreases as you move up the curve, as indicated by the decrease in the value of $\Delta V_F/\Delta I_F$.

Fig 2.10 — Voltage–current relationship for a forward-biased diode

Reading the Forward Curve — Points A, B, C

- **Three Points on the Curve**
  - Point A — zero bias: no voltage, no current
  - Point B — below the knee: VF is less than the 0.7 V barrier; very little current
  - Point C — at the knee: VF ≈ 0.7 V; current begins to rise steeply
- **Above the Knee**
  - VF stays near 0.7 V (up to about 1 V at high current) while IF rises sharply

Dynamic Resistance

- **Resistance That Changes**
  - A diode is non-linear, so it has no single fixed resistance
  - The resistance changes as you move along the V-I curve. This is called the dynamic (AC) resistance
  - Its dynamic (ac) resistance is the slope of the curve: $r'_d = \Delta V_F / \Delta I_F$
- **What It Means**
  - Below the knee the curve is flat → resistance is very high
  - Above the knee the curve is steep → resistance is low and keeps falling as current rises
  - The dynamic resistance of a diode is designated as r'_d

The Reverse V-I Curve & Breakdown

- **The Reverse Side of the Curve**
  - Under reverse bias, only a tiny reverse current flows (typically microamps or less)
  - At the breakdown voltage (VBR) — the knee — reverse current (IR) shoots up sharply
- **How Big Is VBR?**
  - Set by the doping level: a typical rectifier diode exceeds 50 V; some specialized diodes only ~5 V
  - Breakdown is not a normal operating mode for most pn juncition diodes

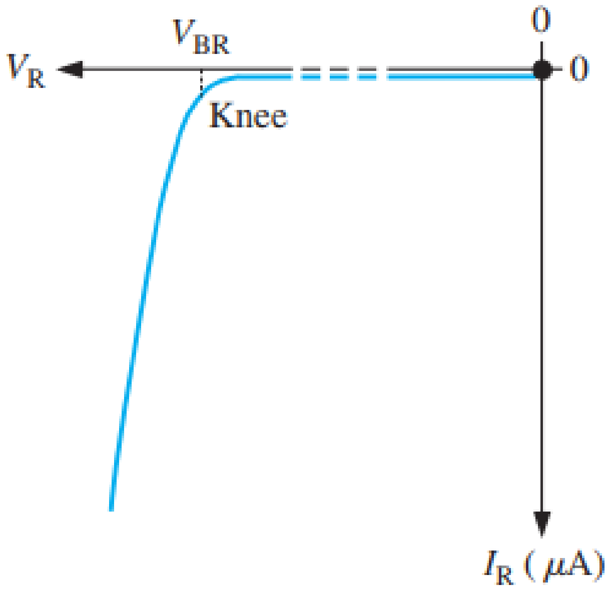


Fig 2.11 — V-I characteristic curve for a reverse-biased diode

The Complete V-I Characteristic Curve

- **Both Halves on One Graph**
  - Combine the forward and reverse curves to capture the diode's entire behaviour in one picture
  - Right of the origin (forward): blocks until the ≈ 0.7 V knee, then conducts strongly
  - Left of the origin (reverse): blocks with only tiny leakage, until the breakdown voltage

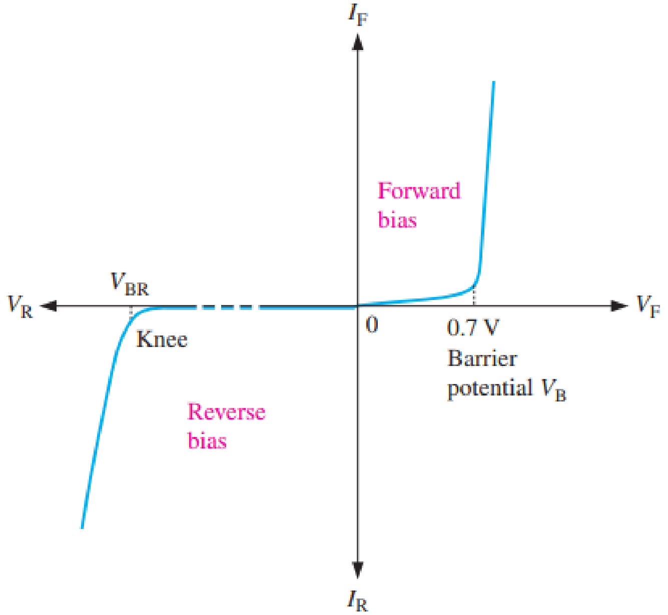


Fig 2.12 — The complete V-I characteristic curve for a diode

Temperature Effects

- **Forward Voltage**
  - The forward voltage decreases as temperature rises — about 2 mV for every 1 °C
  - So a silicon diode drops 0.7 V is really a room-temperature value
- **Reverse Current**
  - Reverse (leakage) current increases with temperature, as more minority carriers are generated
- **Takeaway**
  - Real diode behaviour shifts with temperature — worth remembering for precise designs

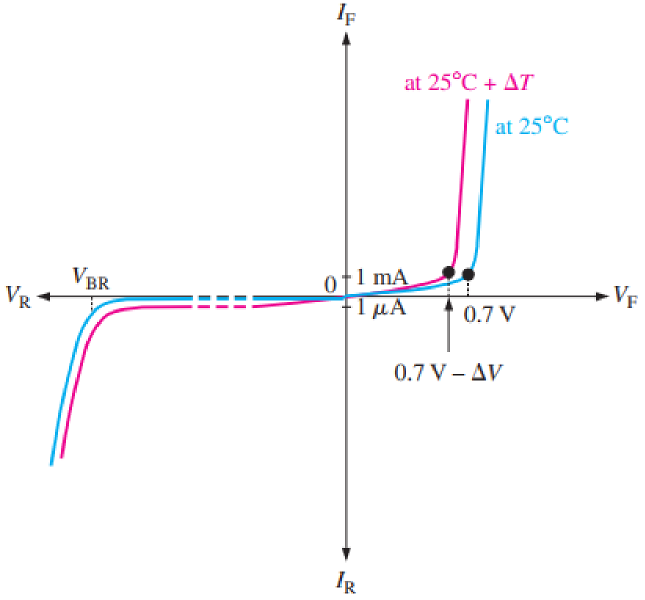


Fig 2.13 — Temperature effect on the diode V-I characteristic

Quick Check #2 — The V-I Characteristic

- **Question 1**
  - What is the 'knee' of the forward characteristic, and roughly what voltage is it for silicon?
- **Question 2**
  - Define dynamic resistance. How does it change as forward current increases?
- **Question 3**
  - Describe the reverse side of the curve up to the breakdown voltage.
- **Question 4**
  - How does forward voltage change with temperature (give the approximate rate)?
- **Question 5**
  - What happens to reverse leakage current as temperature rises?

Part 3 — Diode Models

Why Use Diode Models?

- **The Real Diode Is Complex**
  - Its exact behaviour is non-linear and temperature-dependent
  - For most circuit work we don't need that full complexity
- **Three Levels of Approximation**
  - Ideal model — simplest, for quick analysis and troubleshooting
  - Practical model — adds the barrier potential
  - Complete model — adds resistance, for the most accuracy

Bias Connections — Forward & Reverse

- **Forward-Bias Connection**
  - Connect the positive terminal of the source to the anode (through a current-limiting resistor); the negative terminal of the source to the cathode
  - Forward current (IF) flows from anode to cathode; the forward voltage drop is positive at the anode
- **Reverse-Bias Connection**
  - Connect the negative terminal of the source to the anode; the positive terminal of the source to the cathode (no resistor needed)
  - Reverse current is essentially zero, so the entire bias voltage appears across the diode

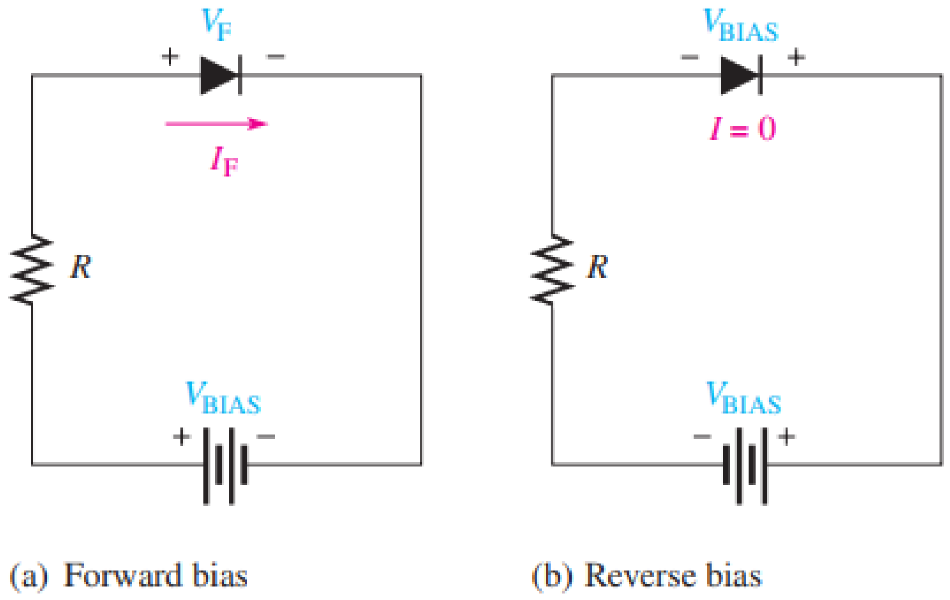


Fig 2.14 — Forward-bias and reverse-bias connections (diode symbol)

The Ideal Diode Model

- **The Diode as a Switch**
  - Forward-biased: a closed (ON) switch — No V across it, current flows
  - Reverse-biased: an open (OFF) switch — no current flows
- **What We Ignore**
  - The barrier potential, forward dynamic resistance, and reverse current are all treated as zero
- **When to Use It**
  - This model is adequate for most troubleshooting when you are trying to determine if the diode is working properly

The Ideal Diode Model Visualized

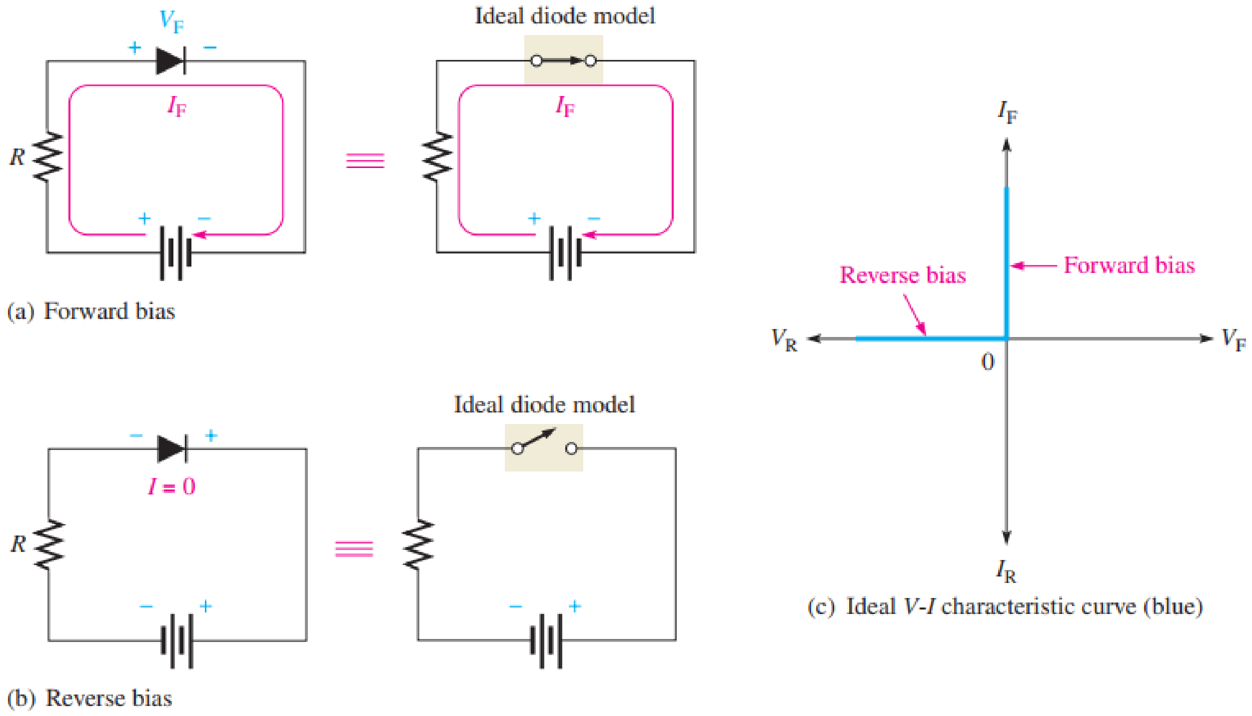


Fig 2.15 — Ideal model: closed switch (fwd), open switch (rev), ideal curve

Ideal Model — V-I Curve & Equations

- **The Ideal V-I Curve**
  - Forward: zero volts across the diode — it lies on the vertical axis, like a plain wire
  - Reverse: zero current — it lies on the negative horizontal axis
- **The Equations**
  - Forward current: Calculated according to Ohm's law

$$V_F = 0\text{ V} \qquad I_F = \frac{V_{BIAS}}{R_{LIMIT}}$$

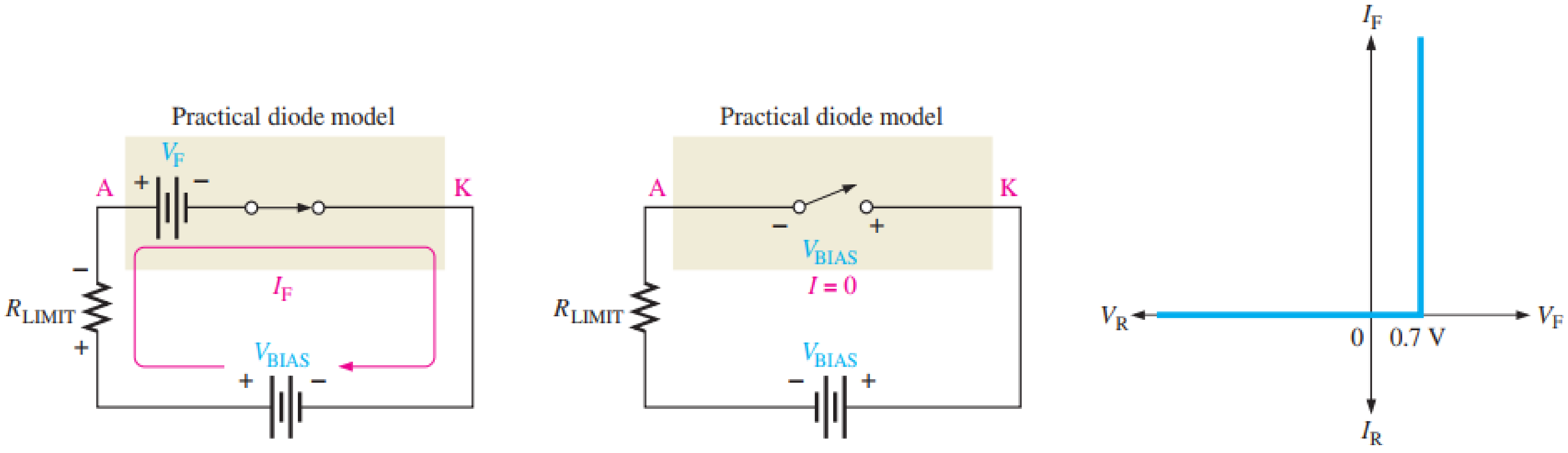
  - Reverse voltage: VR = VBIAS

$$I_R = 0\text{ A} \qquad V_R = V_{BIAS}$$

The Practical Diode Model

- **Add the Barrier Potential**
  - Forward-biased: a closed switch in series with a 0.7 V source (silicon)
  - So the diode drops a constant 0.7 V whenever it conducts
  - Reverse-biased: still an open switch (no current)
- **Why It's Useful**
  - Accounts for the 0.7 V the diode 'uses up' — important when the source voltage isn't large
  - The most commonly used model in practice

The Practical Diode Model Visualized



(a) Forward bias (b) Reverse bias (c) Characteristic curve (silicon)

Fig 2.16 — Practical model: closed switch in series with a 0.7 V source

Practical Model — Forward Current (KVL)

- **Finding the Forward Current**
  - Apply Kirchhoff's voltage law around the loop
  - With VF = 0.7 V:

$$V_{\text{BIAS}} - V_{\text{F}} - V_{R_{\text{LIMIT}}} = 0$$
$$V_{R_{\text{LIMIT}}} = I_{\text{F}} R_{\text{LIMIT}}$$

$$I_{\text{F}} = \frac{V_{\text{BIAS}} - V_{\text{F}}}{R_{\text{LIMIT}}}$$

- **On the Curve**
  - The forward part stands at ≈ 0.7 V; reverse current is taken as zero
  - Best for low-voltage circuits (where 0.7 V matters) and for basic design

$$I_{\text{R}} = 0\ \text{A}$$
$$V_{\text{R}} = V_{\text{BIAS}}$$

The Complete Diode Model

- **On the Forward Side**
  - Add the dynamic resistance: the 0.7 V barrier in series with the dynamic resistance r'd
  - So the diode voltage rises slightly as current increases (not perfectly flat at 0.7 V)
- **On the Reverse Side**
  - A very large reverse internal resistance (r’R) accounts for the tiny leakage current
- **When to Use It**
  - When you need the highest accuracy

The Complete Diode Model Visualized

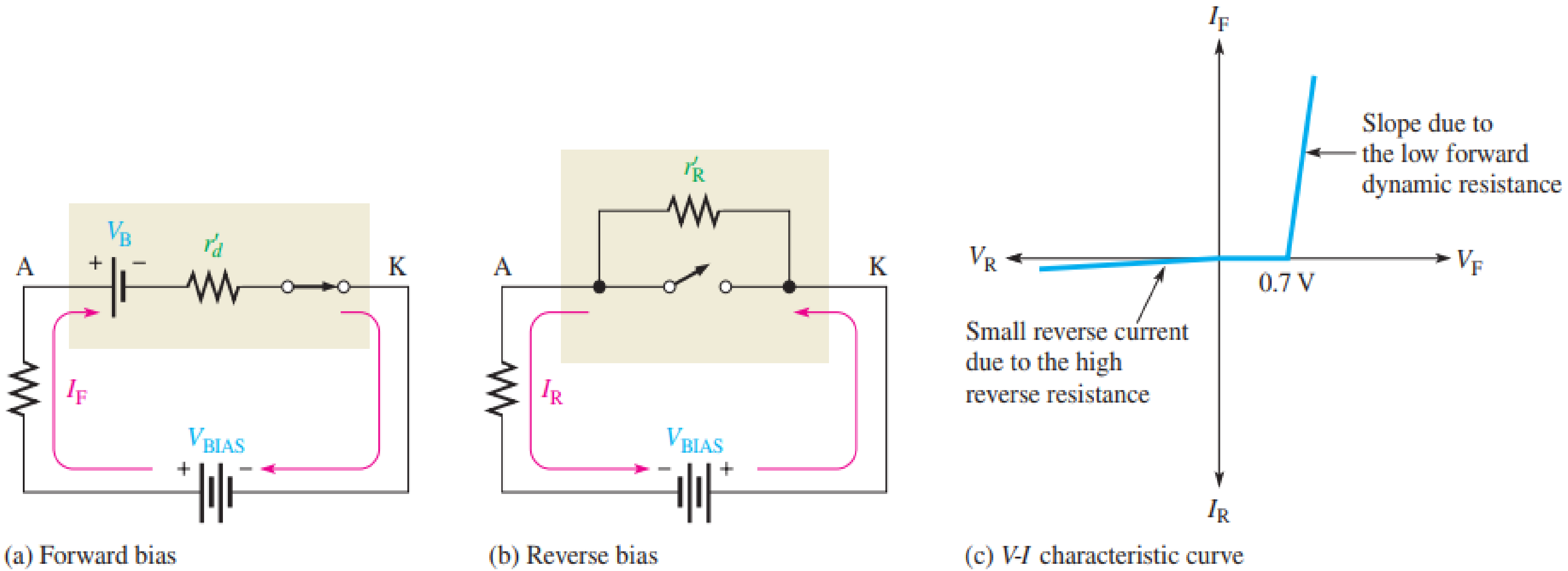


Fig 2.17 — Complete model: barrier + r'd (forward), r'R (reverse)

Complete Model — V-I Curve & Equations

- **Adding the Dynamic Resistance**
  - Forward voltage: VF = 0.7 V + IF·r’d, so VF rises a little as current increases
  - This tilts the forward part of the curve instead of leaving it vertical

$$V_F = 0.7\text{ V} + I_F r'_d$$

$$I_F = \frac{V_{BIAS} - 0.7\text{ V}}{R_{LIMIT} + r'_d}$$

- **Reverse Side**
  - The large internal reverse resistance r’R carries the tiny reverse current
  - Most accurate

The Three Models at a Glance

- Each model adds detail at the cost of simplicity — pick the simplest one that meets your accuracy needs.

Feature	Ideal	Practical	Complete
Barrier potential (0.7 V)	ignored	included	included
Dynamic resistance r’d	ignored	ignored	included
Reverse leakage current	ignored	ignored	included
Diode when ON	wire (0 V)	0.7 V drop	0.7 V + IF·r’d
Best used for	quick checks	most design work	simulation

Worked Example — Forward Bias (Three Models)

- **The Circuit (Fig 2-18a)**
  - (a) Determine the forward voltage and forward current for the diode in Figure 2-18(a) for each of the diode models. Also find the voltage across the limiting resistor in each case. Assume r′d = 10Ω at the determined value of forward current.
  - (b) Determine the reverse voltage and reverse current for the diode in Figure 2–18(b) for each of the diode models. Also find the voltage across the limiting resistor in each case. Assume I_R = 1 μA.

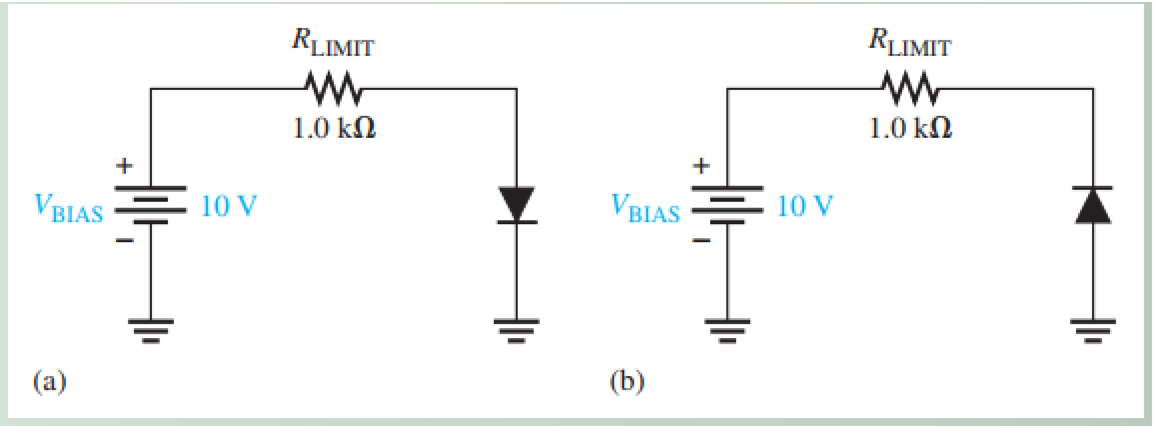


Fig 2.18 — Example 2-1 circuits: (a) forward bias, (b) reverse bias

Worked Example — Forward Bias Solution

- **The Circuit (Fig 2-18a)**

Solution **(a)** Ideal model:

$$V_F = \mathbf{0\ V}$$

$$I_F = \frac{V_{BIAS}}{R_{LIMIT}} = \frac{10\ \text{V}}{1.0\ \text{k}\Omega} = \mathbf{10\ mA}$$

$$V_{R_{LIMIT}} = I_F R_{LIMIT} = (10\ \text{mA})(1.0\ \text{k}\Omega) = \mathbf{10\ V}$$

Practical model:

$$V_F = \mathbf{0.7\ V}$$

$$I_F = \frac{V_{BIAS} - V_F}{R_{LIMIT}} = \frac{10\ \text{V} - 0.7\ \text{V}}{1.0\ \text{k}\Omega} = \frac{9.3\ \text{V}}{1.0\ \text{k}\Omega} = \mathbf{9.3\ mA}$$

$$V_{R_{LIMIT}} = \text{I}_F R_{LIMIT} = (9.3\ \text{mA})(1.0\ \text{k}\Omega) = \mathbf{9.3\ V}$$

Complete model:

$$I_F = \frac{V_{BIAS} - 0.7\ \text{V}}{R_{LIMIT} + r'_d} = \frac{10\ \text{V} - 0.7\ \text{V}}{1.0\ \text{k}\Omega + 10\ \Omega} = \frac{9.3\ \text{V}}{1010\ \Omega} = \mathbf{9.21\ mA}$$

$$V_F = 0.7\ V + I_F r'_d = 0.7\ \text{V} + (9.21\ \text{mA})(10\ \Omega) = \mathbf{792\ mV}$$

$$V_{R_{LIMIT}} = I_F R_{LIMIT} = (9.21\ \text{mA})(1.0\ \text{k}\Omega) = \mathbf{9.21\ V}$$

Worked Example — Reverse Bias Solution

- **The Circuit (Fig 2-18b)**

(b) Ideal model:

$$I_R = \mathbf{0\ A}$$
$$V_R = V_{BIAS} = \mathbf{10\ V}$$
$$V_{R_{LIMIT}} = \mathbf{0\ V}$$

Practical model:

$$I_R = \mathbf{0\ A}$$
$$V_R = V_{BIAS} = \mathbf{10\ V}$$
$$V_{R_{LIMIT}} = \mathbf{0\ V}$$

Complete model:

$$I_R = \mathbf{1\ \mu A}$$
$$V_{R_{LIMIT}} = I_R R_{LIMIT} = (1\ \mu\text{A})\,(1.0\ \text{k}\Omega) = \mathbf{1\ mV}$$
$$V_R = V_{BIAS} - V_{R_{LIMIT}} = 10\ \text{V} - 1\ \text{mV} = \mathbf{9.999\ V}$$

Worked Example — Two Resistors in Series

- **The Circuit**
  - A 12 V source drives a forward-biased silicon diode in series with $R1 = 1.0\ k\Omega$ and $R2 = 560\ \Omega$
  - Find the current and the voltage across each component — practical model ($V_F = 0.7$ V)
- **Solution**
  - Total resistance: $R_T = R1 + R2 = 1.0\ k\Omega + 560\ \Omega = 1.56\ k\Omega$
  - KVL: $I_F = (V_{BIAS} - V_F) / R_T = (12 - 0.7) / 1.56\ k\Omega \approx 7.24$ mA
  - $V_{R1} = I_F \cdot R1 \approx 7.24$ V; $V_{R2} = I_F \cdot R2 \approx 4.05$ V; $V_F = 0.7$ V
  - Check: $7.24 + 4.05 + 0.7 \approx 12$ V ✓ — the diode just adds a fixed 0.7 V to the series circuit

Worked Example — Choosing a Current-Limiting Resistor

- **The Problem (a Design Question)**
  - You want 10 mA of forward current through a silicon diode powered from a 5 V source
  - What series resistor R should you use? — practical model ($V_F = 0.7$ V)
- **Solution**
  - The diode drops 0.7 V, so the resistor must drop the rest: $V_R = V_{BIAS} - V_F = 5 - 0.7 = 4.3$ V
  - Ohm's law: $R = V_R / I_F = 4.3\ \text{V} / 10\ \text{mA} = 430\ \Omega$

Quick Check #3 — Diode Models

- **Question 1**
  - In the ideal model, how is a forward-biased diode treated? A reverse-biased one?
- **Question 2**
  - What does the practical model add to the ideal model?
- **Question 3**
  - What does the complete model add beyond the practical model?
- **Question 4**
  - For a 12 V source and a forward-biased silicon diode, which model is simplest, and why?
- **Question 5**
  - Which model would you choose when the source voltage is only about 1 V? Why?

Part 4 — Wrap-Up

Key Takeaways from Today

- A diode is a single pn junction — anode (p) and cathode (n); the symbol's arrow points with conventional forward current
- Forward bias narrows the depletion region and lowers the barrier → the diode conducts (≈ 0.7 V across silicon)
- Reverse bias widens the depletion region → only tiny leakage current flows, until breakdown (VBR)
- The V-I curve shows the knee (~0.7 V), steep forward conduction, and the reverse/breakdown region
- Forward voltage falls ~2 mV/°C; reverse leakage rises with temperature
- Three models trade simplicity for accuracy: ideal (switch), practical (+0.7 V), complete (+ r'd)
- Use the simplest model that meets your accuracy needs

Coming Up — Week 4: Rectifiers

- **Next Topic — Rectifiers (Lecture Notes pp. 47–65)**
  - Turning AC into DC: the half-wave rectifier
  - Full-wave rectifiers — the center-tapped and bridge configurations
  - Peak inverse voltage and average (DC) output voltage
- **To Prepare**
  - Be comfortable with the practical diode model (0.7 V) — we'll use it constantly
  - Recall that a diode conducts in only one direction
- **Reminder**
  - Read pp. 47–65 before class

Thank You — Questions?